Sepideh Fatemi

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Research Focus

Focuses on scientific foundation models and time-series modeling (forecasting, sparse data, reasoning), scientific discovery (symbolic regression using Large Language Models), and inverse modeling for parameter estimation.

Education

Virginia Tech — Blacksburg, VA, USA
PhD in Computer Science, Advisor: Dr. Anuj Karpatne, **GPA: 4.0/4.0** — 2024-Anticipated May 2028

Virginia Tech — Blacksburg, VA, USA
MS in Computer Science, Advisor: Dr. Anuj Karpatne, **GPA: 4.0/4.0** — 2024-2026

University of Tehran — Tehran, Iran
B.Sc. in Engineering Science - Computer Specialization, **GPA: 3.66/4** — 2017-2022

Research Experience

Virginia Tech, Knowledge Guided Machine Learning Lab — Virginia Tech
Graduate Research Assistant — Jan 2024-present

- ModularSR: Equation Discovery using **Large Language Models (LLMs)** to build a modular Symbolic Regression (SR) framework to enhance interpretability and scalability of modeling complex physical systems.
- LakeFM: **Foundation Model** for aquatic science to, (a) learn effective representations of process variables (b) 2D prediction along temporal and depth axes and (c) continual learning and (d) handling variable cross-frequency learning.
- SAS: Designed an **encoder–decoder** transformer based architecture for time-series prediction with multi-resolution inputs to forecast soil organic carbon and classifying farmer decision type from heterogeneous agricultural data.
- ClaPred: Time sesries prediction using **LSTM**, **Transformer**, and **Transfer Learning** methods, focusing on predicting chlorophyll-a concentrations.
- MCL: Modular Compositional Learning for time series modeling, integrating Machine Learning modules with process-based components.

Convergent Technologies Research Center — Tehran, Iran
Research Assistant — Jan 2022-Aug 2022

- Developed an end-to-end digital phenotyping platform for activity classification using sensory and motor data from mobile devices, comparing individuals with psychiatric disorders to a control group.
- Built a full-stack web service (back-end and front-end) to visualize individuals' activity patterns from GPS data, presented to their healthcare providers at the Atieh Clinic.

Convergent Technologies Research Center — Tehran, Iran
Internship — summer 2021

- Conducted behavioral and neural analysis of human cognitive systems, focusing on memory, judgment, and attention bias to design user studies for story-recall tasks.
- Implemented speech-to-text, Word2Vec, and sentiment analysis tools for NLP tasks involving Persian language text.

Selected Publications

- Abhilash Neog, **Sepideh Fatemi**, Medha Sawhney, Kazi Sajeed Mehrab, "LakeFM: Toward a Foundation Model for Aquatic Ecosystems Using Irregular Multivariate Multi-depth Time Series Data". *KDD 2026, AI4science Track*
- Abhilash Neog, Arka Daw, **Sepideh Fatemi**, Medha Sawhney, et al. "Investigating a Model-Agnostic and Imputation-Free Approach for Irregularly-Sampled Multivariate Time-Series Modeling". *TMLR 2026*
- **Sepideh Fatemi**, Abhilash Neog, Amartya Dutta, M. Sawhney, et al. "Scientific Equation Discovery using Modular Symbolic Regression via Vision-Language Guidance". *CVPR 2025, CV4Science*
- Abhilash Neog, Medha Sawhney, KS Mehrab, **Sepideh Fatemi**, et al. "Toward Scientific Foundation Models for Aquatic Ecosystems". *ICML 2025, FMSD*
- Amartya Dutta, M. Sawhney, K.S. Mehrab, Abhilash Neog, Mridul Khurana, **Sepideh Fatemi**, et al. "Open World Scene Graph Generation using Vision Language Models". *CVPR 2025 CVinW, ICML 2025 Workshop*
- Abhilash Neog, Arka Daw, **Sepideh Fatemi**, Anuj Karpatne. "Masking the Gaps: An Imputation-Free Approach to Time Series Modeling with Missing Data". *NeurIPS 2024, Time Series in the Age of Large Models*
- **Sepideh Fatemi**, Ehsan Maani Miandoab, Nazanin Imandoost, "Study of Nonlinear Dynamics and Chaos in Virotherapy and Radio Virotherapy". *Complex Systems and Network Sciences, 2022*

Teaching Experience

Teaching Assistant – Graduate courses: Machine Learning, Artificial Intelligence — Virginia Tech

Teaching Assistant – Undergraduate courses: Artificial Intelligence, Advanced Programming, Basic Computer Programming, Database Laboratory — University of Tehran

Selected Projects

AI Art Detection: Distinguishing Human-Created from AI-Generated Artworks Fall 2024
- Built a machine learning model using ViT, ResNet-50, and FFT to classify AI vs. human-made art.

Impact of Client Dropouts on Vertical Federated Learning Models Fall 2024
- Investigated the effect of client dropout patterns on the convergence and accuracy of Vertical Federated Learning (VFL) models.
- Implemented neural networks with split learning techniques to analyze feature importance and dropout resilience.
- Evaluated optimizer impact (Adam, RMSprop, AdaGrad, SGD) and experimented with feature sampling methods.

Multi-Agent Reinforcement Learning framework to Analyze Emergence of Social Hierarchies Summer 2023
- Developed a multi-agent reinforcement learning environment using OpenAI's Gym and PettingZoo APIs to simulate resource-constrained scenarios.
- Utilized Deep Q-Networks (DQN) to train agents, observing the natural development of dominance structures and reduced conflict over time.

Professional Activities

Reviewer: KDD 2027, KDD 2026; ICLR 2026 FM4Science Workshop 2026

Lighting talk at CVPR2025, CV4Science 2025
- Talk on "Scientific Equation Discovery using Modular Symbolic Regression via Vision-Language Guidance"

Awarded scholarship for the CMD-IT/ACM Richard Tapia Conference from Virginia Tech 2026

Poster Presentation in 6th Iranian Symposium on Brain Mapping Updates 2022
- Presentation on a smartphone-based digital phenotyping for passive data collection and behavioral analysis.

Member of the Engineering Science Scientific Association, University of Tehran 2018-2019
- Educational and Scientific activities, Certification activities, Competitions and Conferences arrangements

Skills

Programming	Python, C/C++, Java, SQL, Bash, JavaScript
ML Frameworks	PyTorch, TensorFlow/Keras, Scikit-Learn, Hugging Face Transformers, NumPy, Pandas, SciPy
Modeling	Transformers and encoder--decoder architectures, LSTMs, foundation models, transfer learning, LLM/VLM prompting and fine-tuning
Infrastructure & Tooling	Git, Docker, Linux, SLURM/HPC clusters, Weights & Biases, Kubernetes
Data Systems	Postgres, MySQL, MongoDB, Neo4j, ElasticSearch, Kafka
Engineering & Web	MATLAB, Simulink, Flask, Spring Boot, React